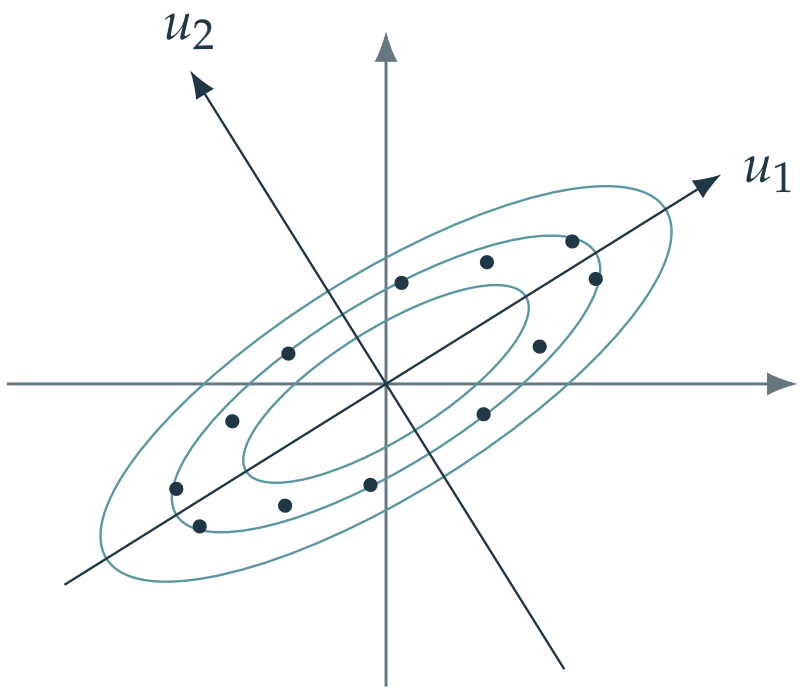
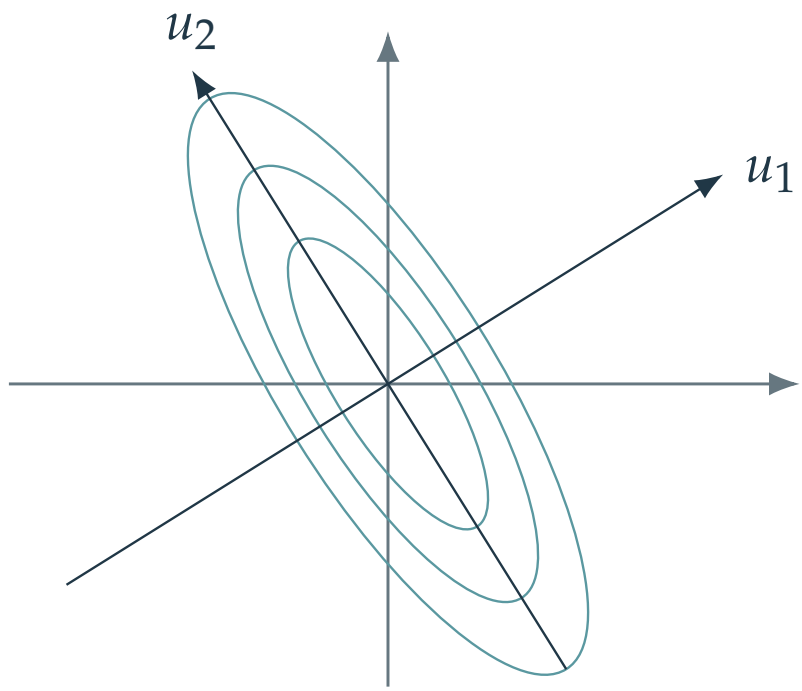
Data covariance

Width along $u_i \propto \sqrt{\lambda_i}$



Quadratic level sets

Width along $u_i \propto 1/\sqrt{\lambda_i}$



$C/n = U\,\mathrm{diag}(\lambda_i)U^\top$

$\langle Cx, x\rangle/n = \text{constant}$

Semiaxes: $r\sqrt{\lambda_i}$ (left), $r\sqrt{\lambda_1\lambda_2}/\sqrt{\lambda_i}$ (right), $r = 1, \frac{3}{2}, 2$.